Extended Relativity:

Space, Time, and the Interpretation Behind Nonlinear Calculus

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Abstract

This note makes explicit an interpretation of relativity that motivated the author's matrix-of-error framework of nonlinear calculus. The interpretation emphasizes space and time as one physical structure and sets out two linked interpretations: an intrinsic density of space affects relativistic behavior, and time exists because of space. Gravitational light-travel delay is its central propagation example: the proposed density supplies a physical reading of metric-dependent travel times while light remains locally measured at c. The time–space statement is an explanation of why relativistic relationships should hold, not a separate new effect established here. The extension connects the resulting matrix-of-error work back to established relativity rather than proposing a replacement theory.

The note connects this account to the shared derivation-and-testing process reported in the preceding nonlinear-calculus paper. That process represented physical corrections as retained matrix contributions and recovered established results across distinct dynamical systems, including three-dimensional Navier–Stokes. Its mathematical results supply a concrete outcome of the interpretation's methodological use, without establishing the interpretation's proposed physical mechanism. The companion density study specifies a dust-plus-vacuum stress and calculates a linear density-to-metric-to-light-delay chain. Those calculations give the density interpretation a concrete candidate realization: the dust-like component can represent a dark-matter contribution, but its calculated response does not distinguish an intrinsic-space interpretation from other sources with the same stress and dynamics. Neither central interpretation is proved by that agreement. The contribution is to connect conceptual motivation, established relativistic mathematics, and the completed calculations without identifying them as the same result. Broader applications are anticipated but are not defined as a new problem class here.

1 Purpose and Contribution

The matrix-of-error framework in *Nonlinear Calculus and Solving Three-Dimensional Navier–Stokes* [1] arose from an interpretation developed before that research program. The guiding idea was that departures from an idealized description can represent physical contributions that should be retained and calculated, rather than treated only as discrepancies. Gravity was a motivating example; nonlinear dynamics supplied a broader setting in which to test the proposed treatment of such contributions.

The interpretation is called *extended relativity* because it extends the explanatory use of the author's reading of relativity, not because it replaces relativity's established predictions. The interpretation

first motivated nonlinear calculus; the resulting derivations and tests can now be related back to that motivation. Its two central claims concern the physical significance of space itself: space has an intrinsic density relevant to relativistic behavior, and time exists because of space. The phrase *if space exists, time exists* summarizes the second claim. Light-travel delay connects these claims to a specific physical phenomenon rather than only to a general statement that spatial and temporal coordinates are related.

This is therefore an interpretation note. Its contribution is the explicit connection between a physical reading of relativity and the matrix-of-error research that it motivated. The preceding nonlinear-calculus paper and companion density study contain the derivations and computations; the present note connects those results to the interpretation without introducing additional tests. Section 2 identifies the established mathematical support. Section 3 states the additional interpretation. Sections 4 and 5 connect it to the framework and the completed density comparison, respectively, including the density realization and light-travel calculation in Section 5.1.

2 The Established Relativistic Setting

Relativity already makes clock readings, spatial measurements, and light propagation parts of a common physical description. Einstein's special-relativity formulation defines distant simultaneity through light signals and transforms spatial and temporal coordinates together [2, Secs. 1–4]. In modern notation, an inertial-frame interval and the proper time of a massive observer satisfy

$$\mathrm{d}s^2 = -c^2\mathrm{d}t^2 + \mathrm{d}\mathbf{x}^2, \qquad \mathrm{d}\tau = \sqrt{1 - \frac{v^2}{c^2}}\,\mathrm{d}t, \tag{2.1}$$

where $v = |\mathrm{d}\mathbf{x}/\mathrm{d}t| < c$. The proper-time reading is attached to the observer's trajectory, not to an observer-independent universal clock.

General relativity extends this description to a spacetime metric. With signature $(-+++)$, $x^0 = ct$, and the cosmological term omitted for this local discussion, the standard equations are

$$\mathrm{d}s^2 = g_{\mu\nu}\,\mathrm{d}x^\mu \mathrm{d}x^\nu, \qquad G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}. \tag{2.2}$$

Here $g_{\mu\nu}$ is the metric, $G_{\mu\nu}$ the Einstein tensor, and $T_{\mu\nu}$ the stress-energy tensor; G is Newton's constant. Geometry and stress-energy enter one dynamical description [3, Secs. 4, 16–17]. This supports treating gravitational effects and measurements together. It does not specify an intrinsic material density of space: the stress-energy tensor is not just one density, and curvature need not vanish in a matter-free region.

A static example makes the clock–light connection explicit. In a region with lapse $N(\mathbf{x}) > 0$, no shift, and time-independent spatial metric h_{ij}, write

$$\mathrm{d}s^2 = -N^2c^2\mathrm{d}t^2 + h_{ij}\,\mathrm{d}x^i\mathrm{d}x^j, \qquad \mathrm{d}\ell^2 = h_{ij}\,\mathrm{d}x^i\mathrm{d}x^j. \tag{2.3}$$

A stationary clock measures $\mathrm{d}\tau = N\,\mathrm{d}t$. Setting $\mathrm{d}s^2 = 0$ for a light ray gives

$$\frac{\mathrm{d}\ell}{\mathrm{d}t} = cN, \qquad \frac{\mathrm{d}\ell}{\mathrm{d}\tau_{\mathrm{local}}} = c. \tag{2.4}$$

The second ratio uses the local stationary observer's clock, not a proper-time clock carried by the ray. Thus a position-dependent relation between coordinate time and local time is compatible

with locally constant light speed. In Schwarzschild's exterior solution, $N^2 = 1 - 2GM/(c^2 r)$ outside the Schwarzschild radius, with M the central mass and r the areal radius [4, Eq. 14]. Equations (2.3)–(2.4) are a direct restatement of standard metric relations, not a new propagation law.

For a light path whose spatial projection is Γ in this static region, Eq. (2.4) integrates to

$$\Delta t_\Gamma = \frac{1}{c} \int_\Gamma \frac{\mathrm{d}\ell}{N(\mathbf{x})}. \tag{2.5}$$

Both the lapse and the spatial metric entering $\mathrm{d}\ell$ matter, and the light path is determined by the spacetime geometry. Coordinate travel time is not itself an invariant local speed measurement. For a return signal timed by one stationary observer, the elapsed clock time is $N(\mathbf{x}_{\mathrm{obs}})\Delta t_{\mathrm{round}}$, where $\Delta t_{\mathrm{round}}$ is the sum of the outward and return coordinate travel times. An observational delay compares such clock readings with a specified reference prediction.

Shapiro proposed precisely such a gravitational radar-delay test: signals passing near the Sun acquire a travel-time contribution relative to the corresponding flat-space calculation [13]. The Cassini radio-link experiment subsequently tested the relativistic propagation effect [14]. This delay is distinct from the ordinary increase of travel time with distance and from propagation through plasma. The established phenomenon is therefore suitable for the present interpretation: gravity changes measured signal timing without requiring a change in locally measured c.

There is also a specific precedent for relating temporal information to spatial geometry. Baierlein, Sharp, and Wheeler outline how two sufficiently nearby three-geometries, subject to Einstein's equations and the conditions of their construction, can determine the intervening spacetime geometry and temporal separation [15]. This is more specific than treating space and time as unrelated inputs. It does not infer time from the bare existence of a single spatial geometry or from a scalar density. A different approach, Connes and Rovelli's thermal-time hypothesis, relates time-flow to a thermodynamic state and an observable algebra [16]. These works provide distinct comparison points, not interchangeable derivations of the author's claim.

3 The Additional Interpretation

The author gives two complementary readings of this common structure: a possible intrinsic origin for a gravitational density contribution, and an explanation of why time belongs to spatial reality. The latter is interpretive context, not a separate propagation law or a new physical effect established by the density calculations.

Intrinsic density of space. Space itself is proposed to possess a nonzero intrinsic mass or density contribution, whose distribution affects gravity, clock behavior, and light propagation. This is a claim about space, not merely about smaller massive objects placed within it. In the motivating picture, massive objects influence this spatial structure, while light propagates through it with locally constant measured speed. Gravitational light-travel delay is the central qualitative expectation: the proposed spatial structure is interpreted as the physical basis of the metric-dependent timing in Eq. (2.5), not as light locally slowing below c. Identifying some gravitationally inferred dark matter with this contribution is a further possible interpretation of the same idea.

Time existing because of space. The author proposes that physical time exists because space exists: temporal behavior belongs to, and depends on, the spatial reality supporting it. The claim concerns physical interdependence, not an equality of the numerical values of space and time or

a maximum lifetime for a finite spatial region. Baierlein–Sharp–Wheeler supplies a mathematical comparison in which spatial geometries carry temporal information [15]; the author's additional assertion concerns why physical time exists at all. The interpretation extends the explanatory reading of that relationship without claiming that the 1962 construction proves this assertion.

These are intended as interpretations of original relativity and were the motivating reason for making this note explicit. The first assigns a possible physical meaning to the structure represented by the metric; the second assigns a proposed dependence to the existence of physical time. Established relativity provides the metric and clock relationships in Section 2. It does not, through those relationships alone, prove either an intrinsic density of space or the asserted reason that time exists.

The observational connection is most direct when the proposed intrinsic contribution is identified with mass otherwise inferred as dark matter. Its distribution would then have to account for gravitational timing, deflection, and clock effects beyond those attributed to visible matter, within a common metric description. The Bullet Cluster lensing analysis, for example, finds a separation between the dominant X-ray-emitting gas and the inferred lensing mass [17]. Such observations constrain the distribution a space-density account must reproduce; they do not by themselves determine whether the underlying contribution consists of particles or has a gravitational origin. If the proposed contribution accounts for mass already assigned to dark matter, it is an alternative identification of that source, not an additional copy of it.

Section 5.1 includes the companion study's specified density-to-metric relation and illustrative linear spatial mode [9, Sec. 7]. The lapse remains part of a metric solution with stress-energy and appropriate geometric data; it is not determined by a local density value alone, and nontrivial propagation effects also occur in vacuum regions. What remains unspecified is an additional relation deriving the density's initial distribution or intrinsic origin. The resulting timing is a calculated consequence of supplied inputs, not an output fitted by choice, but it does not distinguish origins that produce the same stress, dynamics, and metric. Clumping alone would neither identify the contribution as particles nor select mimetic gravity uniquely.

That realization uses a full stress tensor and covariant gravitational coupling, rather than replacing the metric with a scalar density. Its agreement with the matched dust-plus-vacuum description concerns those specified effective dynamics; it does not select their physical origin. Treating light as a probe with negligible backreaction is an approximation, not a claim that electromagnetic energy is absent from the gravitational source [3, Sec. 20]. The candidate's dust four-velocity describes a local flow, not a universal preferred frame, and locally measured light speed remains unchanged. The interpretation's methodological use was to motivate retaining and calculating physical contributions in systems more complicated than their idealized descriptions.

4 Connection to the Matrix-of-Error Framework

Nonlinear calculus was proposed as a shared way to represent and evaluate these contributions across systems. Navier–Stokes was a test of that proposal, not the reason it was originally formed. The organizing expectation was that a system's nonzero corrections could be retained in a matrix and checked against independently established governing behavior [1, Secs. 1–3].

For a system indexed by s, let Q_s be the weak feature matrix formed from its selected physical terms and test functions, and let $\widehat{c}_s$ be the selected coefficient vector. The preceding paper defines

the weighted contribution matrix and its governing-relation residual as

$$\mathfrak{M}_s = Q_s \operatorname{diag}(\widehat{c}_s), \qquad r_s = \mathfrak{M}_s \mathbf{1} = Q_s \widehat{c}_s. \tag{4.1}$$

If K_s indexes the retained correction columns, the matrix of error and their aggregate are

$$\mathfrak{E}_s = \mathfrak{M}_{s,:,K_s}, \qquad a_s = \mathfrak{E}_s \mathbf{1}. \tag{4.2}$$

Here $\mathbf{1}$ is an all-ones vector of the required length. These quantities distinguish zero physical corrections, cancellation among nonzero corrections, and closure of the full governing relation. In particular, $r_s = 0$ need not mean a stationary system. The associated omni-dimensional-equilibrium proposal concerns zero-correction reference cases across systems; it does not identify every balanced evolving solution with stasis.

The word *error* therefore denotes a physical contribution relative to a selected reference description, not a mistake in nature. Calling gravity an error in this terminology means retaining a gravitational contribution that the reference description omitted. The same interpretation can organize other physical corrections without making their particular equations identical. System variables, term libraries, and dimensions differ; the derivation-and-testing process is the shared object.

Weak-form equation recovery has established support in Messenger and Bortz's WSINDy work for ordinary and partial differential equations [5, 6]. SPIDER combines weak formulations, symmetry-informed structure, and sparse identification to recover fluid equations and constraints [7]. These works support the mathematical machinery used in the comparison. The preceding paper's contribution is its cross-system matrix-of-error formulation and tested realizations, not a claim to have originated weak-form recovery. Its reported Mercury, Burgers, and two- and three-dimensional fluid calculations recovered established governing contributions using that shared process [1, Secs. 1–4].

For the unforced incompressible three-dimensional fluid, the pointwise realization is particularly direct. Writing $\omega = \nabla \times u$, $S = (\nabla u + \nabla u^{\mathsf{T}})/2$, and $D/Dt = \partial_t + u \cdot \nabla$, one obtains

$$\frac{D\omega}{Dt} = S\omega + \nu \Delta \omega, \qquad \mathfrak{E}_{\mathrm{NS}} = \begin{bmatrix} S\omega & \nu \Delta \omega \end{bmatrix}. \tag{4.3}$$

Here u is velocity, ω vorticity, S the strain-rate tensor, and $\nu > 0$ kinematic viscosity. Transport is retained in the material derivative. The two columns are stretching and diffusion, whose full block decomposition and numerical realization were derived and tested in [1]. For a smooth periodic solution, their integrated balance is

$$\frac{1}{2} \frac{\mathrm{d}}{\mathrm{d}t} \|\omega\|_{L^2}^2 = \int_\Omega \omega^{\mathsf{T}} S \omega \, \mathrm{d}x - \nu \|\nabla \omega\|_{L^2}^2. \tag{4.4}$$

This connects retained physical contributions to an exact scalar identity. The pointwise columns in (4.3) and their weak tested contributions in (4.1) are different representations, not interchangeable matrix entries.

The demonstrated outcome is consequently specific: the interpretation supported the formation of a framework whose shared process recovered known results and whose retained matrix resolved the contributions required by those systems. The success belongs to those derivations and tests. It provides a practical outcome of the interpretation, rather than a reverse inference that every underlying physical claim must therefore be true.

5 What the Related Investigations Establish

Two comparisons help separate that successful methodological use from a physical explanation of space itself. The first concerns the continuation condition in the nonlinear-calculus paper. Its blockwise estimate connects to the established two-component vorticity regularity criterion of Chae and Choe [8, 1]. The time-integrability requirement arises within the fluid estimates; the conditional smooth-continuation implication can be stated without assuming an intrinsic density of space. That analytic result and the author's broader interpretation have different roles. This note neither supplies an unconditional global-regularity proof nor needs one to explain the framework's conceptual origin.

The second comparison directly concerns gravitationally relevant density. *Gravitational Density: Massive States, Mimetic Matter, and Vacuum Stress* [9], the accompanying study, compares the three descriptions using a common conservation balance. In units $c = 1$, with energy density ρ, pressure p, comoving volume factor $J > 0$, and transfer rate Q, that balance is

$$\frac{\mathrm{d}}{\mathrm{d}t}(J\rho) = -p\dot{J} + JQ. \tag{5.1}$$

Pressureless conserved massive populations give $\rho = \sum_i m_i N_i/J$, where N_i is particle number per reference volume. Mimetic matter gives $\rho = C/J$, with C fixed by its dynamical data [10]. Matching $C = \sum_i m_i N_i$ and the four-velocity matches the effective dust stress tensors. The finite-temperature calculations in [9] quantify an approach to that limit; they do not identify the two descriptions at arbitrary temperature.

Separately conserved vacuum stress instead has $p = -\rho$ and constant ρ [11]. It cannot acquire the same density history solely by being given the same initial density. The Yang–Mills mass-gap question concerns a spectral separation above the vacuum, not the specification of a particle population or the identification of space with dark matter [12]. Mimetic gravity supplies a related effective description through additional dynamical content; it is not ordinary general relativity alone proving that space has intrinsic mass [10].

Those initial comparisons make a conditional effective correspondence and a pressure-dependent distinction explicit. The companion study's added calculation goes beyond homogeneous matching to a specified linear density mode and its gravitational propagation effect [9, Sec. 7]. Its candidate realization and calculated effects are summarized below as further context for the interpretation; no new calculation is performed in this note.

5.1 Density realization and its calculated effects

In units $c = 1$ and signature $(-+++)$, the candidate contribution is

$$T_s^{\mu\nu} = \rho_d u^\mu u^\nu - \rho_v g^{\mu\nu}, \qquad \nabla_\mu T_s^{\mu\nu} = 0, \qquad u^\mu u_\mu = -1, \tag{5.2}$$

where $\rho_d > 0$ is the dust-like density, $\rho_v \geq 0$ is a separately conserved constant vacuum density, and u is the dust four-velocity. The total stress in Eq. (2.2) includes this contribution once. Conservation supplies density transport and geodesic motion. The companion study gives an established constrained-scalar realization from mimetic matter [18, Sec. 2], with its regular potential-flow domain stated explicitly [9, Sec. 7.2]. This is a specified effective candidate, not a new action or a determination of its microscopic origin.

The calculation retains two distinctions. Dust stress does not uniquely select mimetic gravity or identify what carries the density. A homogeneous equation of state also does not by itself

determine perturbation pressure or anisotropic stress. The companion calculation supplies the missing perturbation specification: pressureless dust, no anisotropic stress at linear order, and no perturbation of the constant vacuum component. Its equations then determine the density evolution and metric response from the stated initial and geometric data. With this zero-sound-speed dust specification and matched data, linear agreement with cold dark matter follows from the governing equations, not merely from numerical observation [18]. Potential distinctions concern nonlinear single-flow breakdown or an explicitly modified sound-speed model [18, Sec. 8], neither of which is evaluated in the companion's linear calculation.

The same evolving metric determines light travel. For dimensionless spatial and temporal potentials Φ and Ψ in the weak conformal-Newtonian metric, conformal time η with $\mathrm{d}t = a\,\mathrm{d}\eta$, and background comoving path length ℓ, the first-order propagation contribution is

$$\delta\eta = -\int_\Gamma (\Psi + \Phi)\,\mathrm{d}\ell = -2\int_\Gamma \Phi\,\mathrm{d}\ell, \tag{5.3}$$

where $\Psi = \Phi$ for the specified stress. Unlike the static example in Eq. (2.5), this calculation retains potential evolution. It fixes coordinate endpoints and emission time relative to the corresponding unperturbed background; with zero endpoint potentials, the extra receiver proper interval is $\delta\tau_R = a_R\delta\eta$ to first order, where a_R is the background scale factor at arrival [9, Sec. 7.5]. Local light speed remains c.

In the illustrative mixed background, the comoving density contrast grows by a factor 15.5804 between the initial and reference epochs while the potential amplitude falls to 0.779022 of its initial value. Retaining the potential's evolution during the selected transit gives a delay approximately 0.775% below the frozen-potential reference [9, Secs. 7.4–7.5]. The initial density and background are supplied inputs; these growth and timing values follow from them rather than being selected to match a desired result. They are model calculations, not an observational fit or a new propagation effect beyond general relativity.

The dust-like component can therefore be interpreted as a candidate dark-matter contribution. Mimetic matter already supplies a precedent for a gravitational degree of freedom having effective cold-dark-matter behavior [10]. The author's further reading is that such a contribution belongs intrinsically to space. That identification is not ruled out by the companion calculations, but neither is it selected by them: matched stress, dynamics, and initial/geometric data produce the same response in the tested regular linear regime. The constant vacuum term is distinct and is not being identified as the clustering dust component. No conclusion about an intrinsic origin follows solely from numerical agreement, and the single-mode calculation is not a test of the observed dark-matter distribution. These results supply a concrete example for the intrinsic-density interpretation while leaving that identification open.

The framework anticipates applications beyond the systems already examined because its construction is shared while its inputs are system-specific. No new family of problems, completeness theorem, or computational complexity class is defined here. The anticipation is a consequence of the proposed generality of the framework, not an additional established result. Its support remains the completed cross-system work and the established mathematics to which that work connects.

6 Conclusion

Extended relativity, as stated here, is an interpretation of established relativity organized around two claims: space density affects relativistic behavior, and time exists because of space. Gravitational

light-travel delay gives the first claim a concrete phenomenon to interpret while preserving locally measured c. The Baierlein–Sharp–Wheeler construction provides a specific comparison for the second: spatial geometries can carry temporal information under the field equations, although that does not establish the author's assertion about time's existence.

The concrete contribution of that interpretation was conceptual support for the matrix-of-error framework of nonlinear calculus: physical corrections were to be retained, derived through one shared process, and checked against established outcomes. The preceding paper implemented this direction across distinct systems and developed its full three-dimensional Navier–Stokes realization. Those results are the substantive mathematical outcome connected to the interpretation. The companion density study adds a specified effective stress and a calculated density-to-metric-to-light-delay chain, summarized here in relation to that interpretation. Neither investigation establishes the density's intrinsic origin or the asserted reason that time exists.

The scoped density proposal now has explicit consequences: for the specified stress and initial data, its metric response and light-travel timing agree with the corresponding dust-plus-vacuum description. The dust-like contribution remains interpretable as dark matter whose origin is intrinsic to space; that interpretation is not excluded by these calculations, but the agreement does not distinguish it from other sources with the same effective dynamics. No further relation fixing that origin or its initial distribution is claimed. The time–space statement remains an explanation of why relativity should behave as it does, rather than a separate new effect attributed to the calculation.

The extension consists in carrying this interpretation into a shared mathematical method and relating its completed results back to the relativistic setting that motivated it. It is not a replacement for that setting. Broader applications remain anticipated but unspecified. The note states the interpretation, its methodological outcome, and its calculated gravitational consequences without treating them as interchangeable forms of evidence.

Preparation and Provenance

The author originated the interpretation, the two claims stated in Section 3, and the nonlinear-calculus and omni-dimensional-equilibrium proposals, and directed the research program reported in [1]. This note records that conceptual direction after the mathematical investigations; it does not imply that the interpretation was inferred from those results afterward. Preparation details for the earlier derivations and computations remain in their respective manuscripts. At the author's direction, GPT 6 assembled this first draft, checked the cited primary-source support, and connected the interpretation to the existing manuscripts. GPT 6 subsequently incorporated the companion study's density realization and light-travel results at the author's request, retaining this paper's interpretive purpose and leaving intrinsic origin open. The present note contains no new experimental results.

References

[1] C. M. Struck, *Nonlinear Calculus and Solving Three-Dimensional Navier–Stokes: Cross-System Derivation, Universal Matrix of Error, and Coordinate-Robust Validation*, Zenodo preprint (2026). doi:10.5281/zenodo.22432312.

[2] A. Einstein, "Zur Elektrodynamik bewegter Körper," *Annalen der Physik* **17**, 891–921 (1905). English translation: "On the Electrodynamics of Moving Bodies," *The Principle of Relativity*, University of Calcutta (1920), pp. 1–34. Online English text.

[3] A. Einstein, "Die Grundlage der allgemeinen Relativitätstheorie," *Annalen der Physik* **49**, 769–822 (1916). English translation by S. N. Bose: "The Generalised Principle of Relativity," *The Principle of Relativity*, University of Calcutta (1920), pp. 89–163. Online collection.

[4] K. Schwarzschild, "Über das Gravitationsfeld eines Massenpunktes nach der Einsteinschen Theorie," *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften*, 189–196 (1916). English translation by S. Antoci and A. Loinger, "On the Gravitational Field of a Mass Point according to Einstein's Theory" (1999). arXiv:physics/9905030.

[5] D. A. Messenger and D. M. Bortz, "Weak SINDy: Galerkin-Based Data-Driven Model Selection," *Multiscale Modeling & Simulation* **19**(3), 1474–1497 (2021). doi:10.1137/20M1343166.

[6] D. A. Messenger and D. M. Bortz, "Weak SINDy for partial differential equations," *Journal of Computational Physics* **443**, 110525 (2021). doi:10.1016/j.jcp.2021.110525.

[7] D. R. Gurevich, M. R. Golden, P. A. K. Reinbold, and R. O. Grigoriev, "Learning fluid physics from highly turbulent data using sparse physics-informed discovery of empirical relations (SPIDER)," *Journal of Fluid Mechanics* **996**, A25 (2024). doi:10.1017/jfm.2024.813.

[8] D. Chae and H.-J. Choe, "Regularity of solutions to the Navier–Stokes equation," *Electronic Journal of Differential Equations* **1999**(05), 1–7 (1999). Journal record and article.

[9] C. M. Struck, *Gravitational Density: Massive States, Mimetic Matter, and Vacuum Stress*, version 1, unpublished manuscript, revised 9 September 2026, including Sec. 7.

[10] A. H. Chamseddine and V. Mukhanov, "Mimetic Dark Matter," *Journal of High Energy Physics* **2013**(11), 135 (2013). doi:10.1007/JHEP11(2013)135.

[11] S. Weinberg, "The Cosmological Constant Problem," *Reviews of Modern Physics* **61**, 1–23 (1989). doi:10.1103/RevModPhys.61.1.

[12] A. Jaffe and E. Witten, "Quantum Yang–Mills Theory," in *The Millennium Prize Problems*, J. Carlson, A. Jaffe, and A. Wiles (eds.), Clay Mathematics Institute and American Mathematical Society (2006), pp. 129–152. Official problem description, Sec. 4.

[13] I. I. Shapiro, "Fourth Test of General Relativity," *Physical Review Letters* **13**, 789–791 (1964). doi:10.1103/PhysRevLett.13.789.

[14] B. Bertotti, L. Iess, and P. Tortora, "A test of general relativity using radio links with the Cassini spacecraft," *Nature* **425**, 374–376 (2003). doi:10.1038/nature01997.

[15] R. F. Baierlein, D. H. Sharp, and J. A. Wheeler, "Three-Dimensional Geometry as Carrier of Information about Time," *Physical Review* **126**, 1864–1865 (1962). doi:10.1103/PhysRev.126.1864.

[16] A. Connes and C. Rovelli, "Von Neumann algebra automorphisms and time-thermodynamics relation in generally covariant quantum theories," *Classical and Quantum Gravity* **11**, 2899–2918 (1994). doi:10.1088/0264-9381/11/12/007. arXiv:gr-qc/9406019.

[17] D. Clowe, M. Bradač, A. H. Gonzalez, M. Markevitch, S. W. Randall, C. Jones, and D. Zaritsky, "A Direct Empirical Proof of the Existence of Dark Matter," *The Astrophysical Journal Letters* **648**, L109–L113 (2006). doi:10.1086/508162. arXiv:astro-ph/0608407.

[18] A. H. Chamseddine, V. Mukhanov, and A. Vikman, "Cosmology with Mimetic Matter," *Journal of Cosmology and Astroparticle Physics* **2014**(06), 017 (2014). doi:10.1088/1475-7516/2014/06/017. arXiv:1403.3961; Sec. 2.